

Questions for the written part of the literature seminar

These questions are to be completed in your teams. The articles in Studium serves as a starting point, but please feel free to follow references in those articles and also to look for relevant literature on your own. At least two references in your answers should be to a scientific paper not provided in the Studium. Your answers should contain references to scientific literature, in a consistent referencing format.

Q1: What is the meaning and importance of contextualization and orchestration, and what are currently available tools and services in this area?

Q2: How does contemporary serverless architecture relate to the digitalization of society and Big Data applications?

Q3: Try to, from a technological point of view, relate the following Products/cloud providers, frameworks and tools to each other, both historically and in the technological problems they have addressed. Is there some logical progression in a seemingly fragmented ecosystem?

Set-A: Infrastructure and service related tools and technologies:

OpenStack, Google App Engine, Kubernetes, Ansible, Lambda functions, OpenFaaS

Set-B: Stream processing and model training related frameworks:

Apache Kafka, Apache Pulsar, Apache Spark, TensorFlow, MLFlow, Stackn (of course, include more tools and frameworks if you find things that interests you)

Q4: Explain the role and importance of model serving and CI/CD in today's data engineering world. Write down the names and briefly explain four frameworks (two for model serving and two for CI/CD) that offer model serving and CI/CD capabilities.

Q5: What is the difference between distributed and Federated machine learning? Highlight the architectural differences between the two machine learning environments. Which one of them is more suitable for privacy preserving model training process and why?